



AAE451DBF: A5B

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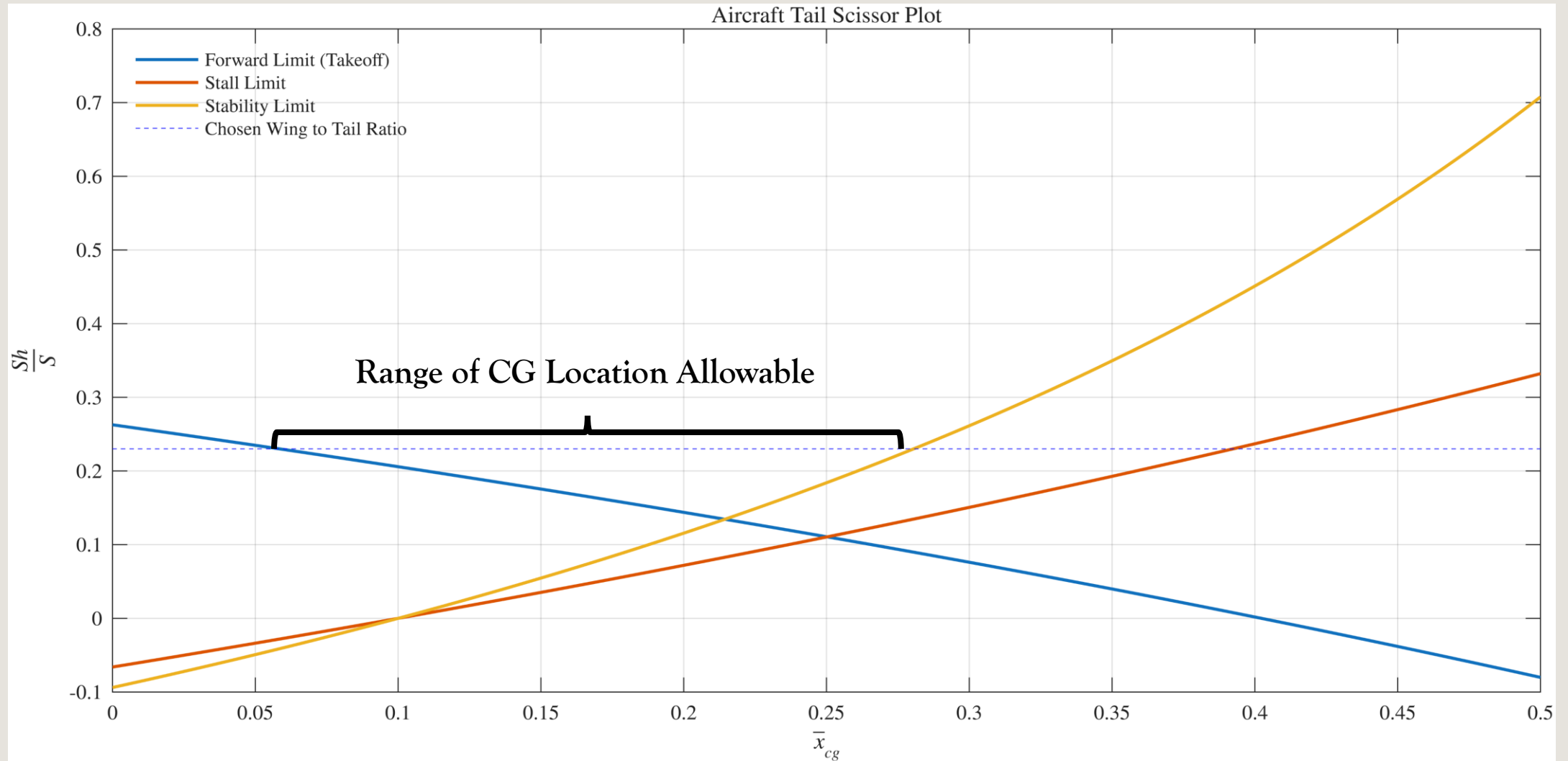
DELIVERABLE 1 – HORIZONTAL TAIL PLANFORM

1. Chosen Airfoil for Horizontal Stabilizer: **NACA 0010**
 - **Justification:** The NACA 0006 and 0008 have really rapid drop offs in the lift curves due to thin airfoil effects, while the NACA 0010 and 0012 do not. The NACA 0012 has higher drag since it is thicker, therefore 0010 is the best choice for this application.
2. Tail Arm – Distance from Wing $\frac{1}{4}$ chord to Horizontal Tail $\frac{1}{4}$ chord: **0.7 m**
3. Horizontal Tail Area: **0.106 m^2**
4. AR: **2.75**
5. Taper Ratio: **1**
6. Elevator Size – As a % of Wing Chord & Wingspan: **61.64%, based off control authority analysis. Will most likely decrease in future if tail moves back.**
7. Additional Tail Parameters Defined in Analysis:
 - Taper: **N/A**
 - Sweep: **N/A**
 - Incidence: **N/A**

DELIVERABLE 2 – SCISSOR PLOT AND CENTER-OF-GRAVITY LIMITS

1. Scissor Plot Assumptions:
 - C_{M_0} is constant over the range of values; $C_{M_0} = -0.08$.
 - Moment coefficient during rotation is 0.1.
 - Elevator Travel Limits is $+10^\circ, -20^\circ$.
 - Max wingspan of 1.5m.
 - Wing Stall Angle of Attack is 15.8° .
 - $C_{L_{MAX}}$ of the Plane is 90% of the Airfoil's C_L .
2. Neutral Point – As a Fraction of Mean Aerodynamic Chord: **0.409**.
3. Horizontal Tail Area: **0.106 m^2** .
4. CG target: **0.259** (% of MAC).
5. Chosen SM: **0.15**.
6. See Scissor Plot in next slide.

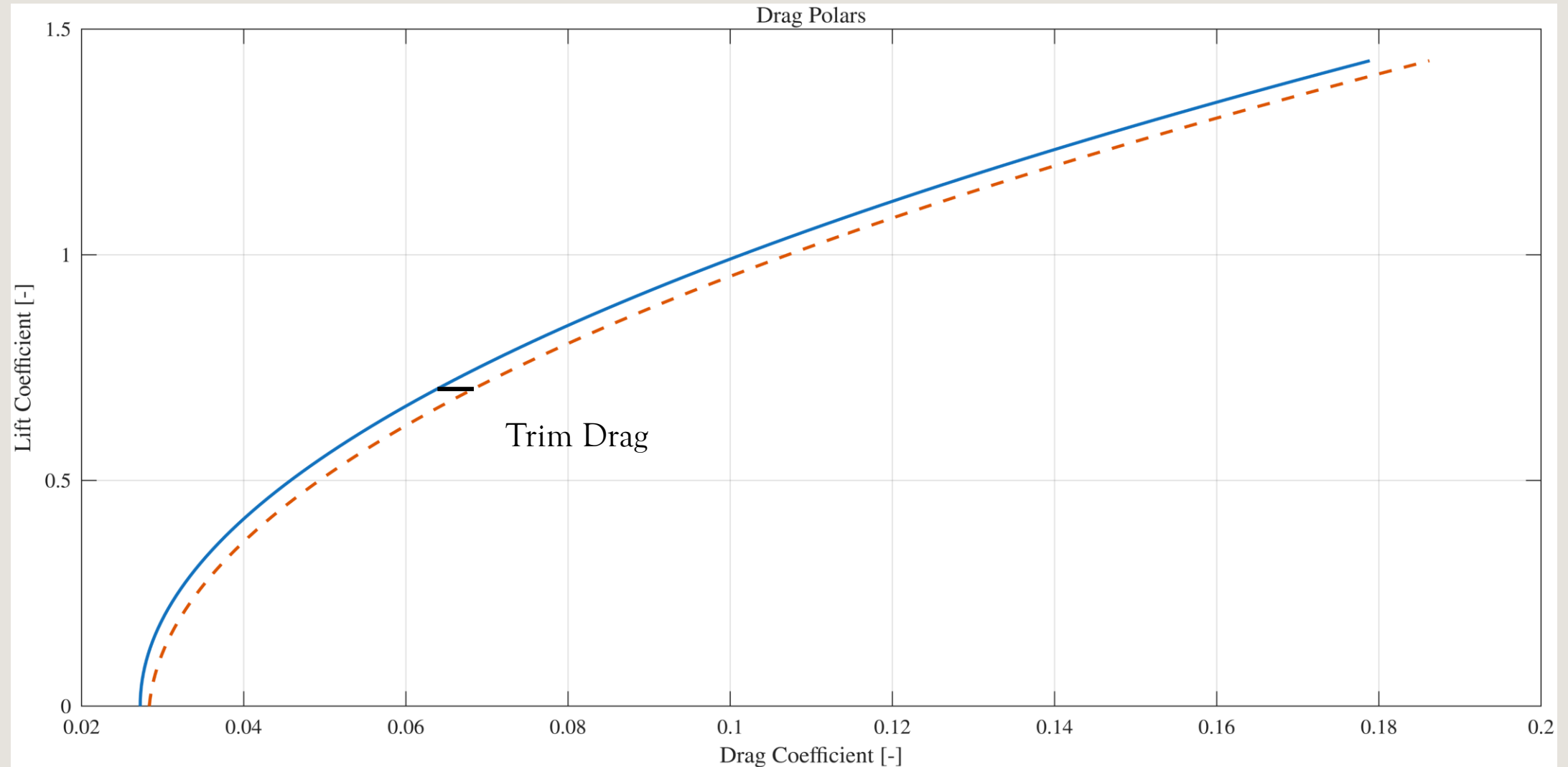
SCISSOR PLOT



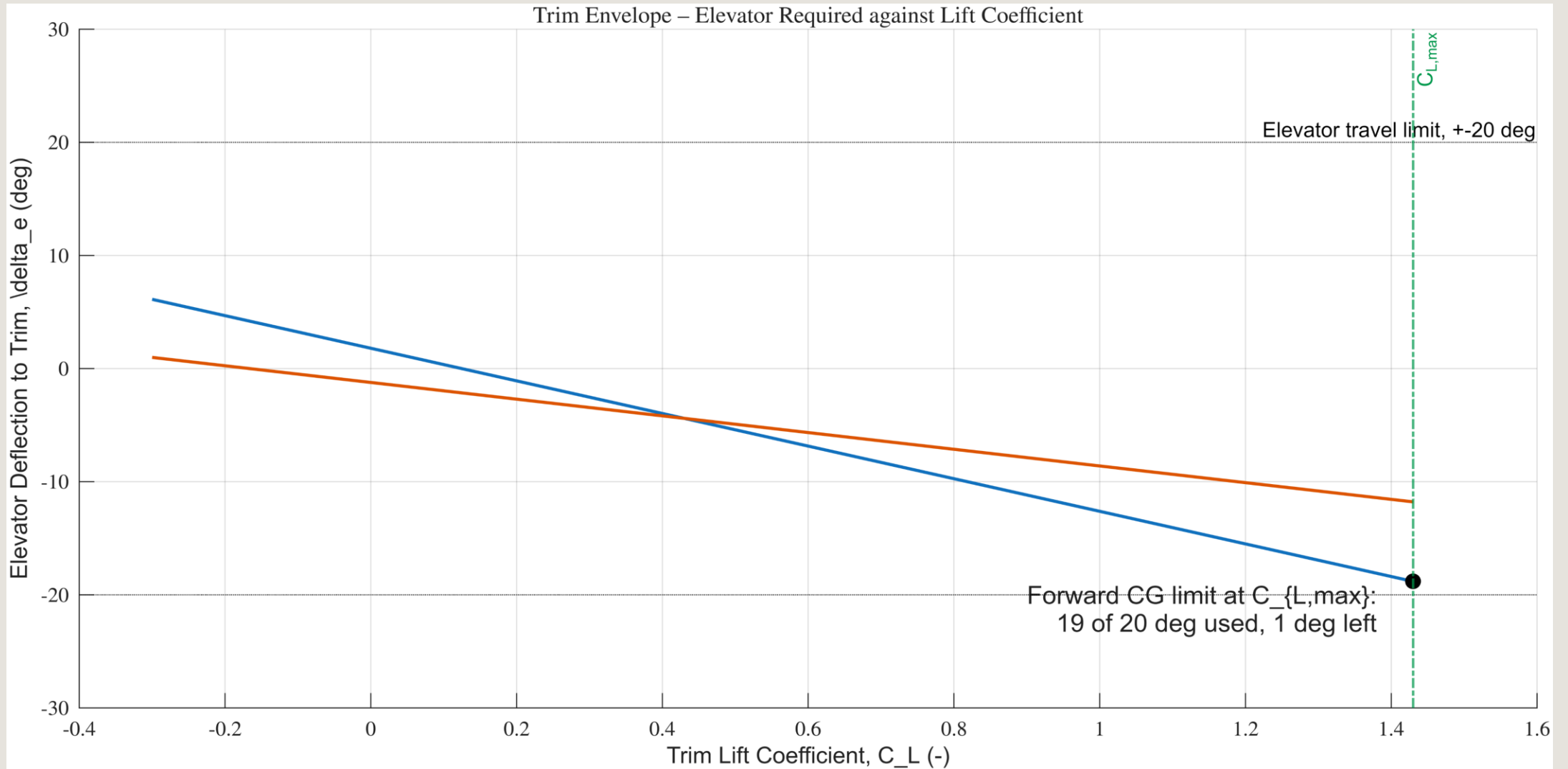
DELIVERABLE 3 – TRIM ANALYSIS

1. Elevator Deflection Required for Trim – Full Flight Envelope @ CG limits: **Worst Case, 1° margin for Foreword Most CG.**
2. See following slides for Drag Polar and Trim Envelope plots.

DRAG POLARS



TRIM ENVELOPE



DELIVERABLE 4 – CONTROL SURFACE SIZING

1. Elevator, Ailerons, and Rudder Sizing – Stating Chord & Span Fraction for Each:
 - **Elevator Chord Fraction: 30%, Span Fraction: 100%.**
 - **Aileron Chord Fraction: 25%, Span Fraction: 70-100% (per side).**
 - **Rudder Chord Fraction: 30%, Span Fraction: 100%.**

2. Authority Demonstration Against Following Criteria:
 - **Elevator – Trim at Forward CG limit, Max Lift Coefficient, with travel margin: 1° of travel margin at Forward CG Limit, assuming 20° control actuation limit.**
 - **Ailerons – Roll rate target at airspeed and source: 120 deg/s, per flight turning speed wanted.**
 - **Rudder – Crosswind / Engine Out Condition, state chosen crosswind ratio: 20% Crosswind Ratio.**

DELIVERABLE 5 – STABILITY DERIVATIVES

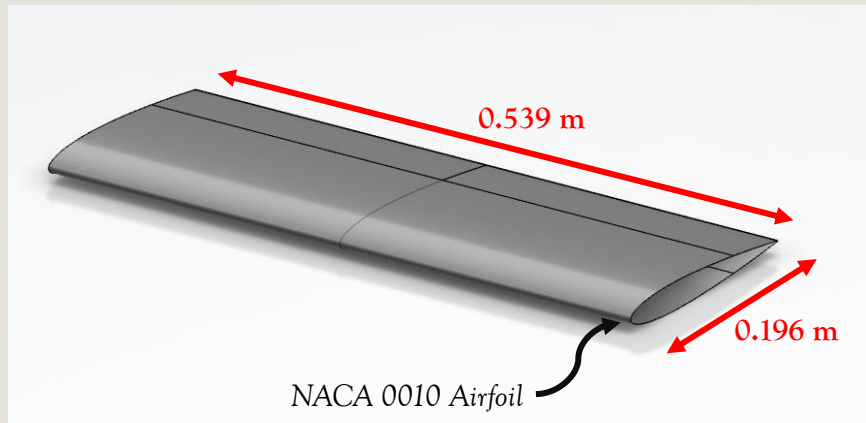
1. Report Longitudinal Derivatives – Lift Curve Slope & Pitching Moment Slope, state CG they were computed at:
 - $C_{M_\alpha} = -0.835 \text{ rad/s}$ (assuming design cg = 0.259).
 - Static Margin: 0.15 (assuming design cg = 0.259).
 - $C_{L_\alpha} = 5.564 \text{ rad/s}$ (assuming design cg = 0.259).

2. Report Lateral-Directional Derivatives – Yawing & Rolling Moment due to sideslip, check against expected range for a stable aircraft of this class:
 - $C_{N_\beta} = 0.1676 \text{ rad/s}$ (expected range 0.05 to 0.15).
 - $C_{L_\beta} = -0.1452 \text{ rad/s}$ at 4° dihedral (expected range -0.05 to -0.2).

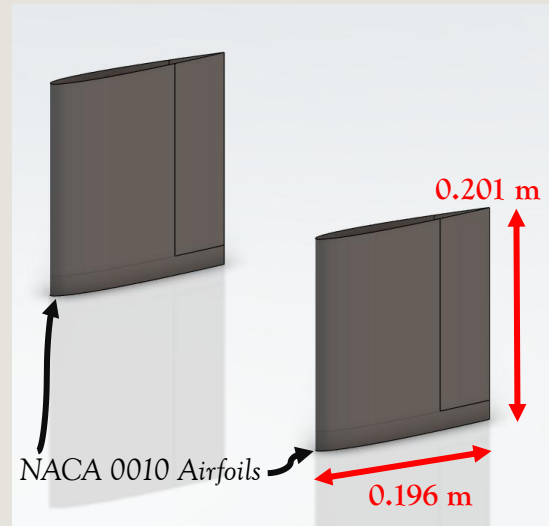
3. Report Dihedral, show effect on Rolling Moment: 4° Up, Effect on Rolling Moment is -0.1359 rad/s.

DELIVERABLE 6 – TAIL MODELS IN 3DX

Updated configuration in team parameterized model is the fifth row in the configuration table in 3DX.



Horizontal Stabilizer Model



Vertical Stabilizers Model



Assembly Model